



Lab 3: Protocols and Encapsulation

Introduction to Protocols and Encapsulation

In computer networks, **protocols** are standardized rules that govern data transmission and reception, ensuring seamless communication between devices regardless of hardware or software differences. Common protocols include:

- **ARP (Address Resolution Protocol)** – Resolves IP addresses to MAC addresses.
- **ICMP (Internet Control Message Protocol)** – Sends messages and operational information.
- **HTTP (Hypertext Transfer Protocol)** – Facilitates web communication.
- **HTTPS (Hypertext Transfer Protocol Secure)** – Secures HTTP traffic with encryption.

Encapsulation is the process of adding protocol-specific headers (and sometimes trailers) to data as it moves down the **OSI (Open Systems Interconnection) model** layers. Each layer appends its own control information, ensuring proper data formatting and transmission across the network. At the receiving end, **decapsulation** occurs, where headers are removed layer by layer to retrieve the original data.

ARP (Address Resolution Protocol)

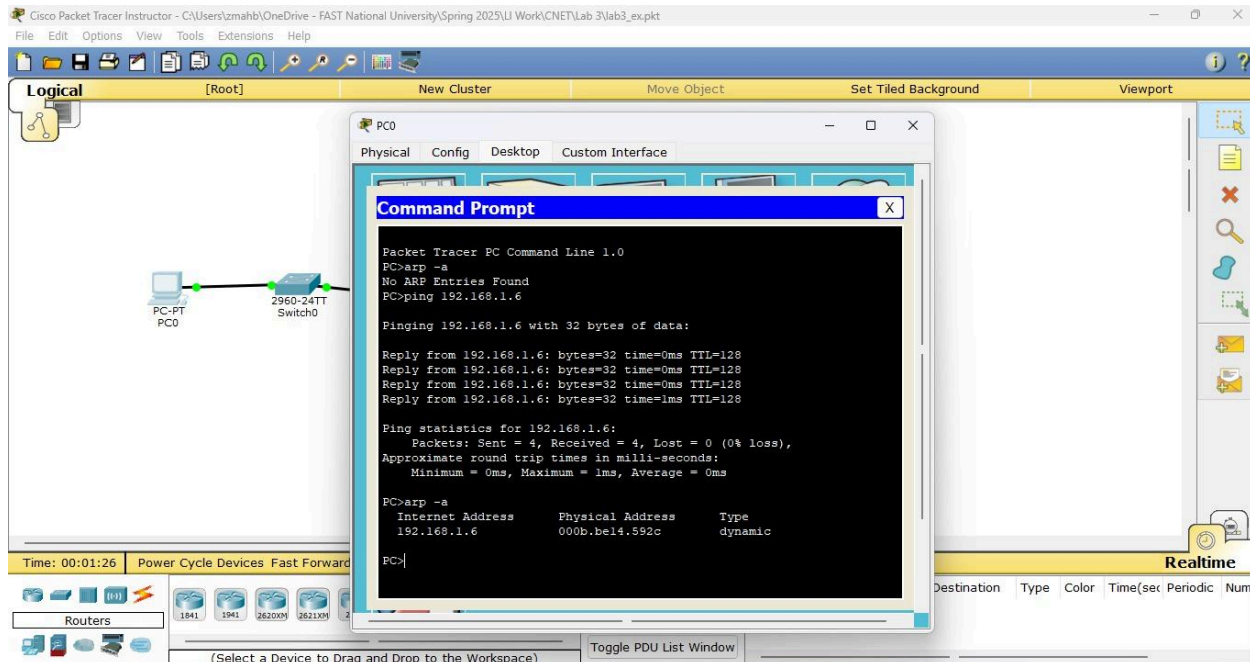
In a network, devices communicate using **IP addresses**, but data is physically transmitted using **MAC addresses**. The **Address Resolution Protocol (ARP)** bridges this gap by mapping an IP address to its corresponding MAC address. This process ensures that packets can be correctly delivered to their destination on a local network.

How ARP Works

1. When a device needs to send data to another device on the same network, it checks its **ARP table** to see if it already has the destination's MAC address.
2. If the MAC address is not found, the sender broadcasts an **ARP request** to the network, asking, *"Who has this IP address?"*
3. The device with the matching IP responds with an **ARP reply**, providing its MAC address.
4. The sender stores this information in its ARP table for future use, avoiding the need for repeated requests.

ARP in CISCO Packet Tracer

To perform an ARP simulation in Cisco Packet Tracer, start by creating a simple network with two PCs (PC1 and PC2) connected to a switch using Copper Straight-Through cables. Assign IP addresses to the PCs: **192.168.1.1** for PC1 and **192.168.1.2** for PC2.



Open the **Command Prompt** on PC1 by navigating to **Desktop > Command Prompt**, and enter `arp -a` to check the ARP table, which should initially be empty. Next, send a **ping** request from PC1 to PC2 by typing `ping 192.168.1.2`.

Since PC1 does not yet know PC2's MAC address, it first sends an **ARP request** to the network, and PC2 responds with its **ARP reply**, providing its MAC address (you can see this in motion by sending packets through simulation mode). Once the MAC address is resolved, the ping is successfully sent. After the ping completes, run `arp -a` again to observe the updated ARP table, which now contains PC2's MAC address.

ICMP (Internet Control Message Protocol)

The **Internet Control Message Protocol (ICMP)** is a fundamental protocol used for network diagnostics and error reporting. It operates at the **Network Layer (Layer 3)** of the **OSI model** and is primarily used to test connectivity, report network errors, and help manage network performance. Unlike TCP or UDP, ICMP does not carry user data but instead provides feedback on network conditions.

How ICMP Works

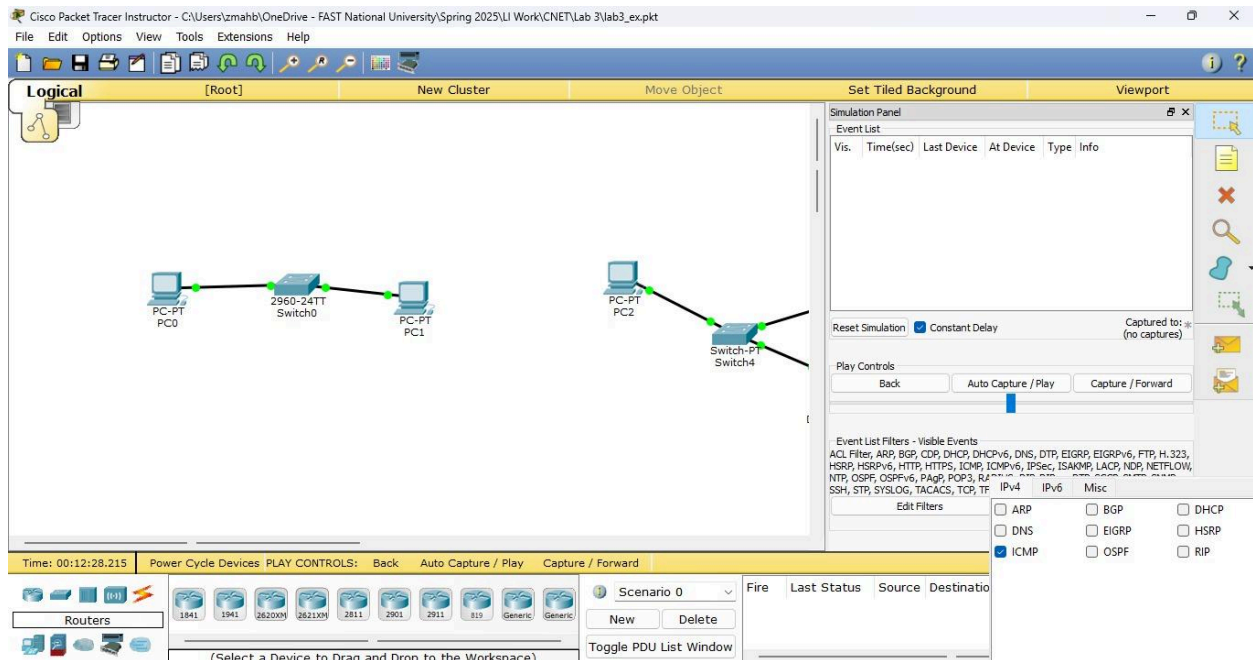
1. **ICMP Echo Requests and Replies:** The most common use of ICMP is the **ping** command, which sends an **echo request** to a destination device. If the destination is reachable, it responds with an **echo reply**.



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2. **Error Reporting:** If a packet cannot be delivered (e.g., due to a destination being unreachable or a timeout), ICMP generates an error message to inform the sender.
3. **Network Troubleshooting:** ICMP helps diagnose latency, packet loss, and connectivity issues using tools like **ping** and **traceroute**.

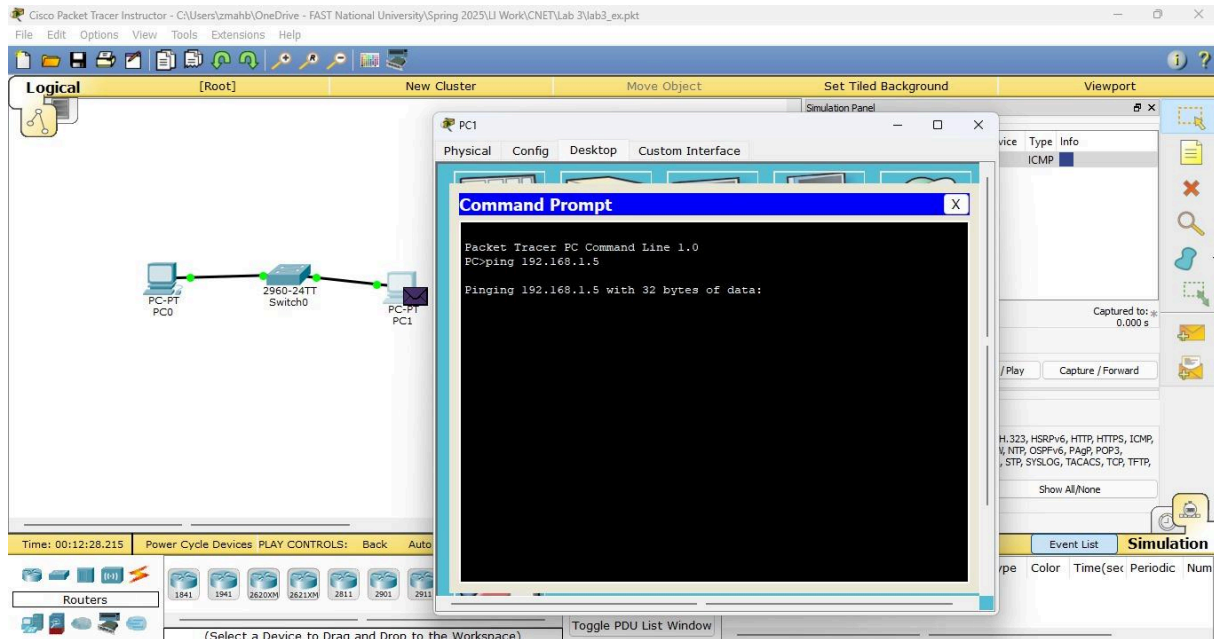
To perform an ICMP simulation in Cisco Packet Tracer, first, set up a network with two PCs (PC1 and PC2) connected to a switch using Copper Straight-Through cables, and assign them IP addresses: **192.168.1.1** for PC1 and **192.168.1.2** for PC2.



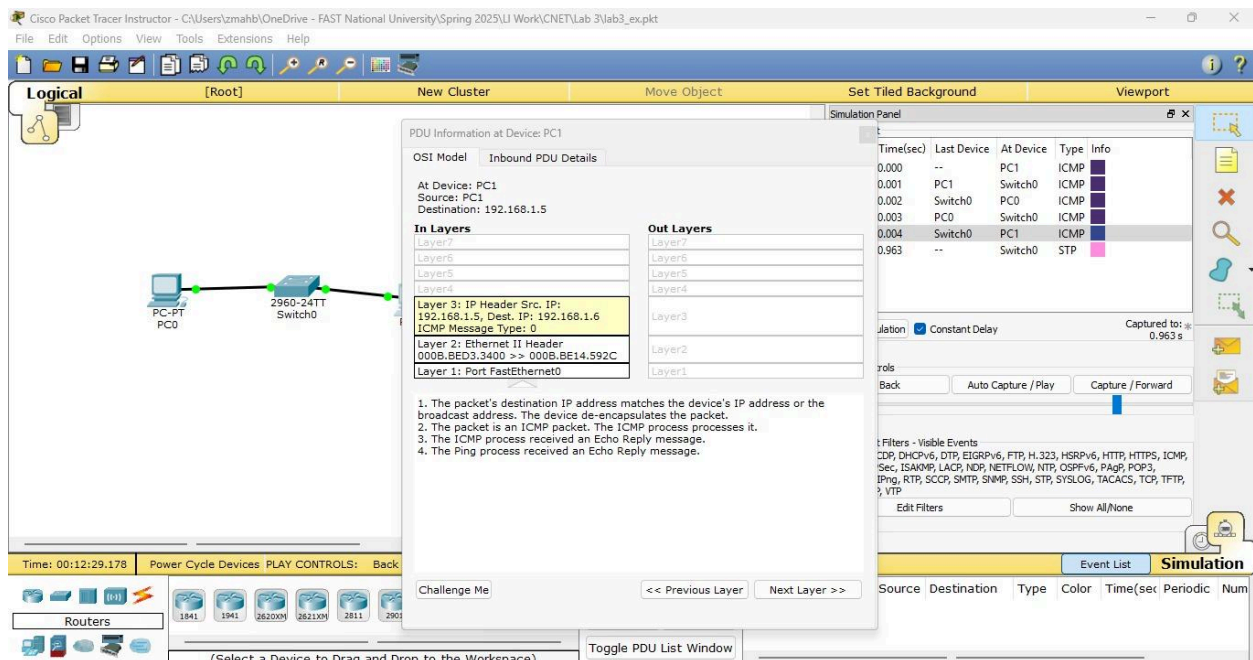
Open **Simulation Mode** by clicking the **Simulation button** at the bottom-right corner and configure the **Event List Filters** to display only **ICMP** packets.



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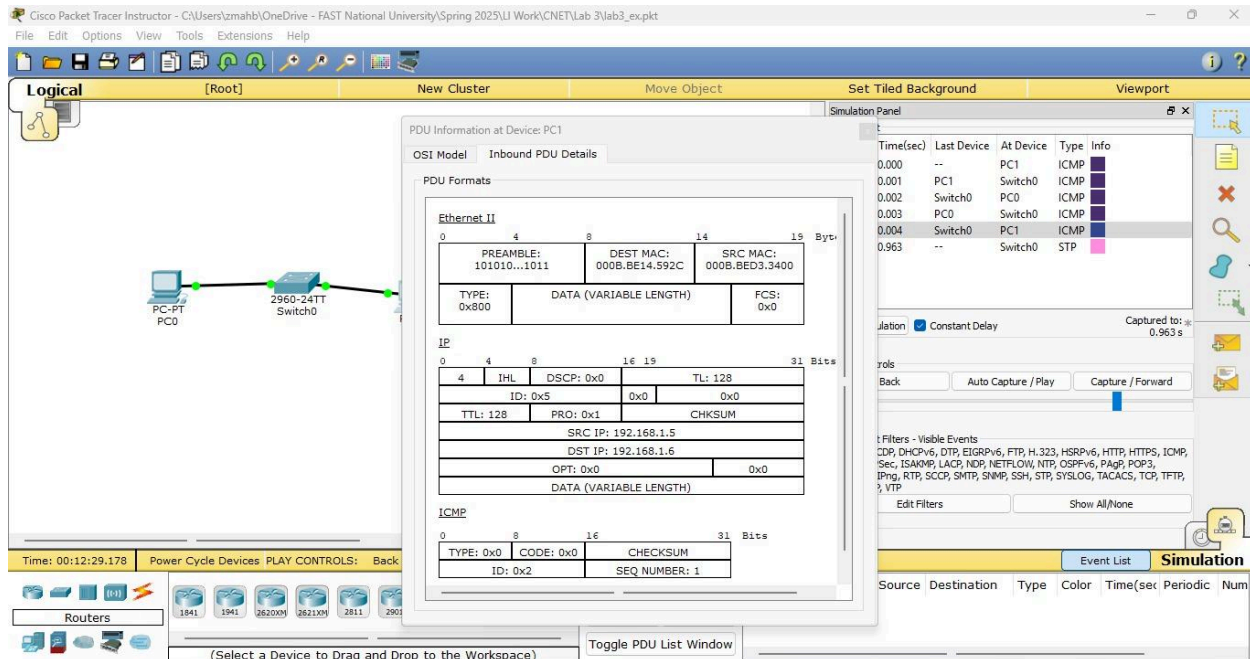
On PC1, open the **Command Prompt** by navigating to **Desktop > Command Prompt** and enter ping 192.168.1.2 to send an ICMP Echo Request to PC2.



Observe the **ICMP packets** as they move through the network. Click on a packet in the Event List to inspect details such as the **source and destination IP addresses** and the **ICMP**



message type (Echo Request or Echo Reply).



After PC2 responds, verify that the **ICMP Echo Reply** returns to PC1, confirming successful communication.

HTTP and HTTPS

HTTP (Hypertext Transfer Protocol) is a foundational protocol for the web, enabling the transfer of hypertext (web pages) between a web server and a client (browser). However, HTTP transmits data in plain text, making it vulnerable to interception. **HTTPS (Hypertext Transfer Protocol Secure)** addresses this vulnerability by encrypting data using **SSL/TLS** protocols, ensuring secure communication between the server and client. HTTPS protects the integrity and confidentiality of data, providing a secure browsing experience, especially for sensitive transactions like online banking or shopping.

Key Differences

- **HTTP:** Transfers data in plain text, susceptible to eavesdropping and man-in-the-middle attacks.
- **HTTPS:** Encrypts data, safeguarding it from interception and tampering, and verifies the authenticity of the website via certificates.

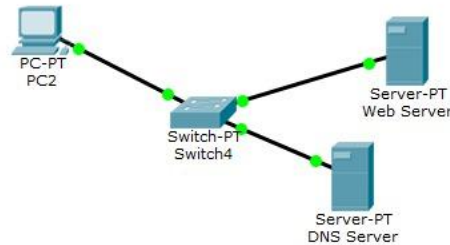
HTTP/HTTPS in CISCO Packet Tracer

To perform HTTP and HTTPS encapsulation in Cisco Packet Tracer, first, create a network



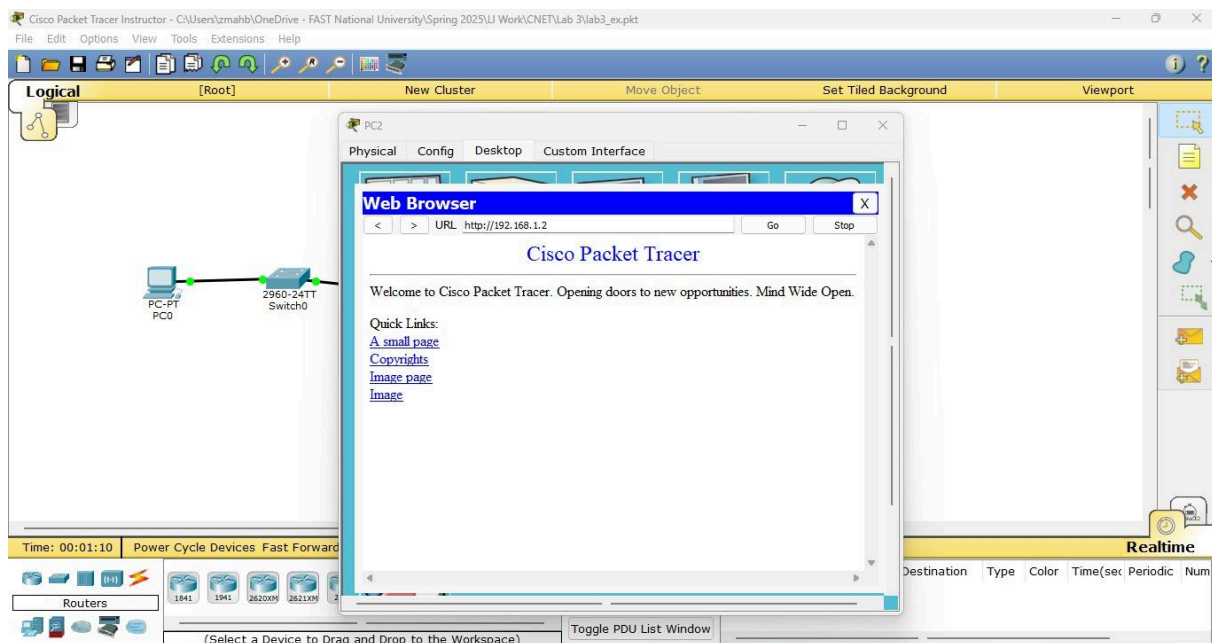
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topology consisting of one PC (Client), one Web Server, one DNS Server, and a Switch to interconnect them. Assign the following IP addresses: PC (192.168.1.10), Web Server (192.168.1.100), and DNS Server (192.168.1.1). Use Copper Straight-Through cables to connect all devices.



Network Topology to simulate HTTP/HTTPS protocol

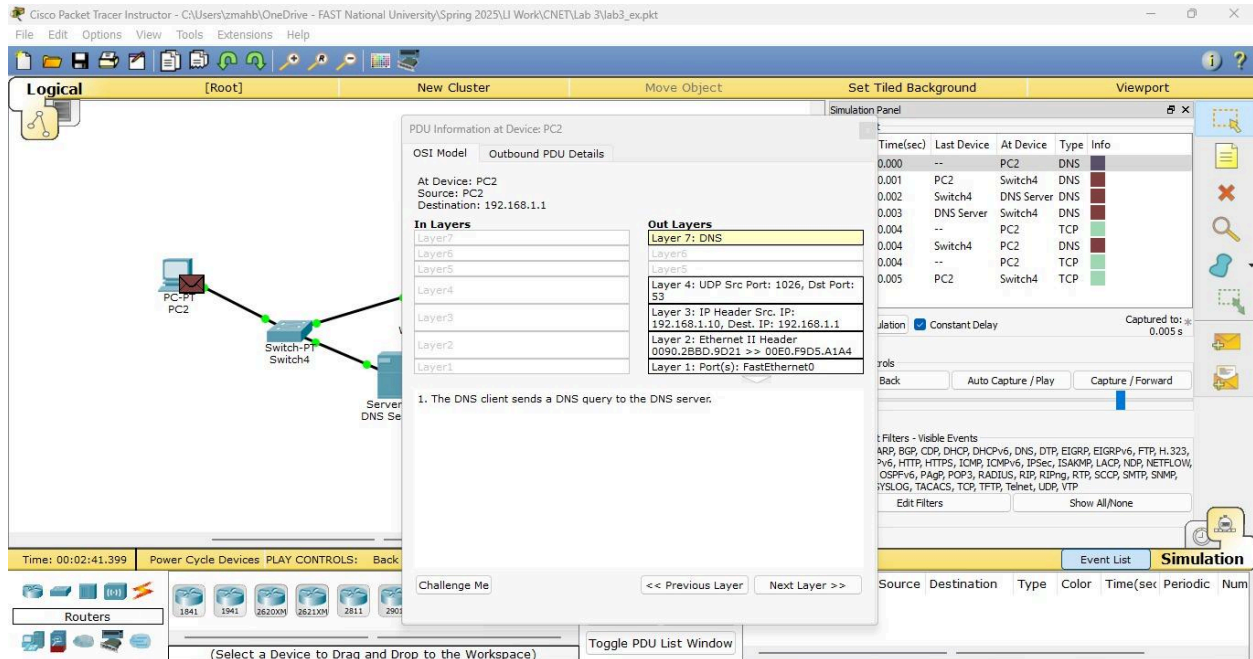
Next, configure the Web Server by enabling HTTP and HTTPS services under the Services tab. Then, set up the DNS Server by enabling the DNS service on your DNS Server and adding an entry for "webserver.com" pointing to 192.168.1.100. On the PC, navigate to **Desktop > IP Configuration** and set the DNS Server field to 192.168.1.1. Open the web browser on the PC and access the Web Server by entering "http://webserver.com" or the IP and you will be able to see a website.





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Switch to Simulation Mode, and refresh the browser to observe the HTTP request traveling to the Web Server and the response returning to the PC. Click on the packet to analyze the details of the HTTP GET request and response.



Sending HTTP/HTTPS packets in simulation mode.

Lab Tasks

Note: All tasks must be documented in a single report. Include screenshots of every step (description is not as necessary as the screenshots).

CLO2: Apply core networking concepts and protocols, including but not limited to routing, Congestion control, and error management, to establish and analyze reliable data communication across networks.

Task 1: Simulating ARP Operation (5 Marks)

Open Cisco Packet Tracer and create a simple network topology with two PCs connected to a switch (2960). Assign IP addresses to the devices, with PC1 using 192.168.1.1 and PC2 using 192.168.1.2. On PC1, open the command prompt and use the `arp -a` command to display the empty ARP table. Next, use the `ping` command on PC1 to send a request to PC2.

After executing the ping, check the ARP table again with `arp -a`. This time, the table should contain an entry for PC2's IP address along with its corresponding MAC address. Also, capture screenshots of the ARP table before and after the ping to document the changes.



Finally, write a brief explanation of how ARP functions to map IP addresses to MAC addresses, ensuring that devices on a network can communicate efficiently.

Task 2: ICMP Simulation (5 Marks)

Using the same network from Task 1, ensure that both PCs are properly connected with their assigned IP addresses. Open the Simulation Mode in Packet Tracer to observe network activity in real time. From PC1, initiate a ping to PC2 by entering ping `192.168.1.2` in the command prompt. As the packets travel between the two devices, observe the ICMP messages being exchanged. Click on the packets to inspect their details within the Simulation Mode to provide visual evidence of the communication process. Make sure to take screenshots of the following packets:

1. ICMP Echo Request
2. ICMP Echo Reply

Also mention the difference between an ICMP Echo Request and an ICMP Echo Reply.

Task 3: HTTP Encapsulation (5 Marks)

Set up a web server and client to inspect HTTP request and response packets. In Cisco Packet Tracer, create a network consisting of a PC, a web server, and a DNS server, all connected via a switch (**Switch PT**). Assign the IP addresses as follows: PC (192.168.1.10), Web Server (192.168.1.100), and DNS Server (192.168.1.1). Use Copper Straight-Through cables to establish connectivity between the devices.

Configure the web server by enabling HTTP and HTTPS services under the Services tab. Then, configure the DNS server by enabling the DNS service and creating an entry for "lab3-[yourRollNumber].com" that maps to 192.168.1.100. In the Web Server, edit the index.html page and add a heading 1 tag which contains your name, roll number, section and lab number. On the PC, navigate to the Desktop > IP Configuration section and set the DNS server address to 192.168.1.1.

Open the web browser on the PC and access the web server using "lab3-[yourRollNumber].com". Switch to Simulation Mode then refresh the browser to observe the HTTP request being sent to the server and the response being returned. Click on the packets to analyze details such as the HTTP GET request and response headers. Take screenshots of:

1. Accessing the website by the domain name (lab3[yourRollNumber].com).
2. Accessing the website by its IP address.
3. Packet details for an HTTP Request happening in layer 7.



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4. Packet details of the packet received by the PC which helps it display the page in the web browser. (**hint**: look for a packet that has HTTP reply written on it and happens on the PC)

Also provide a brief description of what is happening in the packet for parts 3 and 4.

Submission Guidelines (5 Marks):

- Submit a single PDF document containing all tasks and the CISCO packet file(s) to the GCR, **ensure your file name is visible in all pictures and contains your roll number**.
- Ensure that all screenshots are clear and labeled appropriately.
- Include your name, roll number, and lab number inside the **first page** of the document.
- Name the submitted document as follows: SectionLetter_RollNumber_LabNumber, for example: **L_i231234_Lab1**.
- Name the CISCO files as follows: SectionLetter_RollNumber_TaskNumber, for example: **L_i231234_Lab1**. (**for multiple files add the task number at the end**)
- The unethical usage of Generative AI is strictly prohibited and will result in 0 marks.
- **In case a demo is conducted, there will be deductions based on your answers to the questions asked.**

NOTE: Screenshots where your name, roll number, section AND lab number are NOT visible, will automatically be given 0/not considered valid.

Sample screenshot:

